

# LUKAS RETSCHMEIER

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I am a Software Engineer with more than seven years of professional experience across multiple domains. My responsibilities have included technical leadership, solution design, and full-stack software development. I have extensive experience with modern software development practices, tools, and architectures.

I hold a PhD in algorithms, with research interests at the intersection of data science and data privacy. My work has been published at leading computer science and machine learning venues, reflecting a strong track record of developing rigorous solutions to challenging technical problems. I particularly enjoy working on problems that combine an interesting mathematical theory with meaningful practical impact.

**Technologies:** Java + Spring, AngularJS, Docker | *Additionally:* Python, C++

## EDUCATION

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**Doctor of Philosophy (PhD) in Algorithms** 23 - 26  
University of Copenhagen Supervisor: *Prof. Rasmus Pagh*  
*Focus on Differential Privacy*

Research stay abroad at Boston University

**Master of Science in Computer Science** 19 - 23  
Technical University of Munich Supervisor: *Paloma Thomé de Lima*  
Exchange semesters at Universitetet i Bergen and IT University Copenhagen

**Bachelor of Science in Computer Science** 15 - 18  
Cooperative State University Mannheim, ECTS Grade A Supervisor: *Prof. Karl Stroetmann*  
*in cooperation with Atos Information Technology and Siemens Professional Education*

## PROFESSIONAL EXPERIENCE

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**Atos Information Technology**, PLM Consultant, *Selected Projects:* 15 - 23

**Home Appliances Industry** — Technical Lead [Blog Post](#) 21 - 23  
Led the development of a system for real-time monitoring and error detection of thousands of production orders by streaming and consolidating production-line data across heterogeneous databases and factories. 100+ internal users per month, including key decision-makers.  
Technologies: Full-stack development, *Java* (with *Spring*) as backend, *AngularJS* front-end – microservice-based, SQL

**Automotive Industry** — PLM Consultant 19 - 21  
Developed key components of an after-sales platform, with a focus on implementing a toolchain for validating, converting, and rendering media files.  
Technologies: *Java*, *Docker*, *Siemens Teamcenter*, *IBM MQ*

**Automotive Industry** — Junior PLM Consultant 18 - 19  
Containerized Teamcenter on Linux and used it to evaluate the error-prone process of directly upgrading Teamcenter from version 10 to 12. Significant time-savings because containerization allowed us to quickly try again with other parameters.

Technologies: *Java, Docker, C++, Siemens Teamcenter*

**Additional responsibilities:** mentoring of students, support of bid processes, experimental PoCs (e.g. Teamcenter containerization, in the cloud)

### Academic Employment

Teaching: Discrete Structures (21), Algorithms and Data Structures (23, 24, 25)

**External Examiner** for computer science, Denmark

26 – present

## LANGUAGES & CERTIFICATIONS

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**German** (native); **English** (C1); **Danish** (C1, Studieprøven); **Norwegian** (intermediate, B1)

ITIL®3 Foundation (APMG), Business English Level 2 (LCCI)

## PUBLICATIONS

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- [1] Jacob Imola, Rasmus Pagh, and **Lukas Retschmeier**.  
“*Lower Bounds for Private Graph Problems via Reconstruction Attacks*”. In: PODS 2027.
- [2] Lukas Geis, Alexander Leonhardt, Johannes Meintrup, Ulrich Meyer, Manuel Penschuck, and **LR**.  
“*Revisiting a Successful Reduction Rule for Dominating Set*”. In: ALENEX 2026.
- [3] Rasmus Pagh, **Lukas Retschmeier**, Hao Wu, and Hanwen Zhang.  
“*Optimal Bounds for Private Minimum Spanning Trees via Input Perturbation*”. In: PODS 2025.  
—**Note:** Improves *Faster Private Minimum Spanning Trees* (TPDP 2025) by subset of authors.
- [4] Joel Daniel Andersson, **Lukas Retschmeier**, Rasmus Pagh, and Boel Nelson.  
“*Private Lossless Multiple Release*”. In: ICML 2025.
- [5] Christian Janos Lebeda and **Lukas Retschmeier**.  
“*The Correlated Gaussian Sparse Histogram Mechanism*”. In: FORC 2025.

### Under Submission

- [6] Anders Aamand, Rasmus Pagh, and **Lukas Retschmeier**.  
“*Optimal Bounds for Differentially Private All Pairs Shortest Paths*”.
- [7] Paloma Lima and **Lukas Retschmeier**.  
“*A Linear Kernel for Planar Semitotal Dominating Set*”.

## COMMUNITY SERVICE

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**Reviewer** — ICML (25), NeurIPS (26), TMLR (25, 26), RANDOM (25) — SEA (26)

**PC Member** — TPDP (26), SatML (26)

## HONORS

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Best graduate in informatics (*DHBW Mannheim*)

A first prize in the piano competition *Jugend Musiziert*

## OTHER INTERESTS

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Piano and classical music, (winter) bathing, volleyball